

Exploring a Safe-Driving Instruction Framework for Older Drivers Based on the GDE Framework and Motor Learning Theory: A Pilot Study

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Abstract

This pilot study examined the relationship between driving instructors' approaches and risky driving behaviors among older drivers in three stages: driving school familiarization, on-road driving with behind-the-wheel training (BWT), and on-road driving with minimal intervention (MI). Two frameworks—goals for driver education (GDE) and internal-external focus—classified instructor utterances. Cluster analysis was applied to high-frequency words, and two-way ANOVA tests showed significant differences in instructional categorizations and stages for both frameworks. Correlation analyses indicated that under the BWT stage, riskier behaviors were inversely related to internal and internal $\times$ external instructions. Conversely, these behaviors were positively correlated with external, internal $\times$ external instructions, and GDE-based instructions in the MI stage, including the total number of instructor utterances. These results suggest instructors compensate for reduced physical intervention by increasing verbal instructions, emphasizing the need for adaptive training strategies, and the possibilities of external guidance elements and future advanced driver assistance systems incorporating internal.

CCS Concepts

• **Human-centered computing** $\rightarrow$ **Accessibility technologies.**

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Keywords

Older drivers, Driving Instruction, Goals for driver education, Internal-external focus

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1 Introduction

Road safety is influenced by a complex interplay of factors including driver familiarity with vehicle operations, traffic conditions, regulation adherence, and the ability to anticipate and respond to emergencies [6][15][22]. While obtaining a driver's license establishes basic driving skills and safety awareness, significant risks persist. Experienced drivers may develop unsafe habits over time [9], and older drivers often face cognitive and physical declines that elevate crash risk and the risk of death [28][12][1][2]. Consequently, numerous studies advocate for remedial training programs for traffic violators and targeted reassessments for older drivers [16][4] to reinforce safe driving practices and enhance road safety.

Hence, the goal for driver education (GDE) framework has gained recognition as a robust theoretical and practical model for driver instruction [6][22]. The framework organizes instructions into four hierarchical levels: (1) vehicle maneuvering, (2) mastery of traffic situations, (3) driving goals and contexts, and (4) goals for life and skills for living [6]. Previous studies have focused primarily on novice drivers, particularly in-vehicle coaching and instruction [15][22]; however, less attention has been devoted to older drivers, who may also require adaptive instructional approaches. Understanding how GDE-informed strategies affect driving behaviors



Figure 1: Viewing the vehicle and driver as a unified system, vehicle maneuvering aligns with an internal focus, while mastery of traffic situations corresponds to an external focus.

in older adults could help develop tailored and age-appropriate interventions.

Recent technological developments advanced driver assistance systems (ADAS), highlight their potential to enhance driving safety [10, 11]. Voice-guided ADAS can deliver real-time, context-sensitive feedback during critical moments [4][21]. However, existing research has largely overlooked how GDE-based instructional methods might be adapted for automated, voice-driven systems. Additionally, some studies exclude the “goals for life and skills for living” category from real-world instruction because it does not directly influence moment-to-moment driving decisions [22][23]. It suggests that a simplified GDE framework, informed by motor learning theory, may be more effective for real-time applications. One promising approach involves categorizing driver guidance into internal (body/vehicle-focused) and external (effect of the movement/environmental-focused) attentional focus [20] [17][25]. Viewing the driver-vehicle system as an integrated entity enables alignment between vehicle maneuvering and internal focus and mastery of traffic situations with external focus (Figure 1).

Although many studies have explored novice driver training, relatively few have focused on the unique needs of older adults [15]. Concurrently, ADAS technologies have emerged as pivotal advancements in human-computer interaction (HCI), providing real-time feedback and voice-guided support that can also provide driving-related instruction. However, the need for ADAS assistance to reduce the potential risks associated with physical decline is even greater for older drivers. Research indicates that for individuals over the age of 50, acceptance of the ADAS pivots on factors such as ease of use, alignment with social norms, and trust in technology [24][3]. Older adults, especially those with cognitive or psychological limitations, may require more flexible visual and auditory alerts as well as reduced system complexity [16]. However, existing ADAS implementations sometimes remain too complicated or intentionally introduce unnecessary barriers [24][3]. Such design choices can undermine older adults’ confidence in the ADAS and impede broader adoption. To address these challenges, this pilot study analyzed real-world in-vehicle instructions provided to older drivers, aiming to evaluate the applicability of a structured GDE framework and a simplified internal-external (IE) focus framework in designing user-centered ADAS systems. Specifically, this study has three objectives: (1) to investigate how the GDE framework

is used during practical driving sessions with older adults, (2) to evaluate the feasibility of adapting the GDE framework into an IE focus framework, and (3) to assess correlations between these frameworks and risky driving behaviors. By integrating insights gathered from real-world instruction and established theoretical models, this study aims to guide the design of simplified ADAS voice guidance systems that not only enhance safety and usability for older adults but also improve the effects of ADAS for a diverse and expanding driving population.

2 Related work

2.1 GDE framework and older driver education

Empirical evidence suggests that applying the GDE framework enhances risk perception and decision-making among novice drivers [6][15]. However, limited research has explored its adaptation for retraining older drivers [16][4]. Existing studies often prioritize classroom-based or written assessments rather than on-road interventions tailored to the cognitive and physical requirements of older adults. Investigating the natural integration of GDE elements in real-world instructional settings can provide valuable insights into designing more effective coaching strategies for these demographics.

2.2 Speech strategies and IE focus

Verbal communication influences drivers by providing feedback, encouragement, and directives. Previous research indicates that a supportive yet authoritative tone can effectively prompt drivers to rectify unsafe practices [27][19]. However, most studies have focused on novice or general populations, overlooking the specific needs of older drivers who may face challenges in attention, motor skills, or information processing. Research suggests that adopting an external focus, emphasizing the effects of movement with the environment, often results in superior motor performance compared to an internal focus, which focuses on body movements. Nevertheless, the extent to which driving instructors encourage older drivers to adopt a particular attentional focus remains unclear. This study examined whether internal and external focus types are implicitly incorporated in real-world instructor–driver interactions to mitigate unsafe behaviors among older drivers.

2.3 ADAS voice interfaces and driver compliance

ADAS are crucial tools for enhancing driver safety [4][21]. Voice-based ADAS can issue real-time alerts regarding hazards or speed compliance, encouraging safer driving practices. Driver acceptance of these systems depends on factors such as altered frequency, tone, and perceived system reliability [7]. However, research on the unique responses of older drivers to ADAS voice strategies remains limited. Moreover, the potential for transferring speech patterns used by instructors to automated guidance systems remains underexplored. This study aims to inform the development of adaptive, age-specific ADAS voice systems by analyzing how the GDE framework or a simplified IE focus framework, appears in practical driving instruction.

3 Method

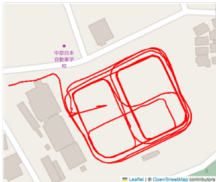
3.1 Participants and experiment

This study included 24 older drivers (mean age = 66.5 ± 8.7 years; 12 male, 12 female) and four experienced male driving instructors (mean age = 45.0 ± 2.2 years; mean instructional experience = 19.5 ± 2.1 years). Supplementary Material 1 provides detailed participant characteristics and instructor–driver assignments. Ethical approval was granted by the relevant ethics committee.

A dual-brake experimental vehicle was used, with the primary brake located in the footwell of the driver and the secondary brake installed in the front passenger seat, enabling the instructor to intervene when necessary. The experiment was divided into three distinct stages, conducted over two consecutive days:

- Stage 1 (Day 1, Driving school familiarization): Older drivers completed several laps within the driving school grounds (Figure 2a) until the instructor determined they were sufficiently familiar with the experimental vehicle, a process lasting 10–15 min.
- Stage 2 (Day 1, On-road behind the wheel training (BWT)): Drivers navigated urban and residential streets (Figure 2b), starting at position 1, driving to the target area 2, and completing 6–10 loops before returning to 1. Instructors provided navigation and coaching during this stage, using the passenger-side brake when necessary.

a. Driver school practice area



b. Urban residential streets

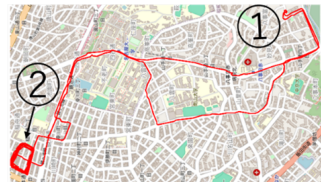


Figure 2: (a) Practice area designated for initial training at the driving school. (b) The urban route is used for on-road training, starting at 1 (driving school), proceeding to 2 (target area), and completing six counterclockwise laps before returning to 1.

- Stage 3 (Day 2, On-road driving minimal intervention (MI)): Drivers repeated the same route under similar conditions. However, instructors minimized direct intervention and focused primarily on providing navigation cues.

A BU-DRHD421 driving recorder (Yupiteru Corporation) with GPS and acceleration sensors continuously captured vehicle data throughout all stages while its integrated voice recorder documented all verbal instructions. However, some data were lost during file import and export procedures using Yupiteru's PC Browser (version 1.9.37.0). Additional details regarding the missing data are outlined in supplementary 1.

3.2 Data processing and analysis

Audio recordings were transcribed into Japanese text using the Notta Transcription Service. Word frequency analysis and extraction were conducted using the KH Coder 3.00 (official package) software [8], with the top 93 words subjected to cluster analysis. We referred to a prior study [22] to align these terms with GDE-related categories and classified them into four domains of driving behavior: (1) vehicle maneuvering, (2) mastery of traffic situations, (3) goals and contexts for driving, and (4) navigation (supplementary 2). After that, the frequency of instructor utterances containing words from each category was calculated. Utterances containing at least one term from a specific category were assigned to that category, while those without relevant terms were labeled unclassifiable (GDE). This process produced five GDE-related utterance categories: (1) vehicle maneuvering, (2) mastery of traffic situations, (3) driving goals and contexts, (4) navigation, and (5) unclassifiable (GDE). Frequent words were mapped to either focus type (supplementary 2) to investigate the influence of internal versus external focus of attention on driving behavior. Utterances that did not align with either focus were labeled unclassifiable (IE). This mapping generated four IE focus categories: (1) internal only, (2) internal $\times$ external, (3) external only, and (4) unclassifiable (IE). Each utterance was exclusively assigned to one of these categories to prevent overlap. Proportions of utterances for each category were calculated relative to the total number of utterances at each stage, considering session length variations (e.g., the number of red lights encountered).

Aggressive driving behaviors were assessed using established metrics, including maximum driving speed, counts of unsafe acceleration ($> +0.3g$) [14], rapid braking ($< -0.3g$) [26], and unsafe turns (lateral acceleration $> 0.3g$) [11], where g represents standard gravitational acceleration (9.81 m/s^2).

3.3 Statistical analysis

Two-way ANOVAs were conducted with stages and utterance categories as factors within the GDE and IE frameworks to evaluate whether the proportions of instructional utterances differed across stages and categories in these classifications. Post hoc pairwise comparisons were conducted based on ANOVA results. Correlations between instructor utterance counts and aggressive driving behavior (maximum speed, unsafe acceleration, rapid braking, and unsafe turns) during on-road driving were analyzed to identify potential relationships. Statistical significance was set at $p = 0.05$. All analyses were conducted using Python 3.11.7.

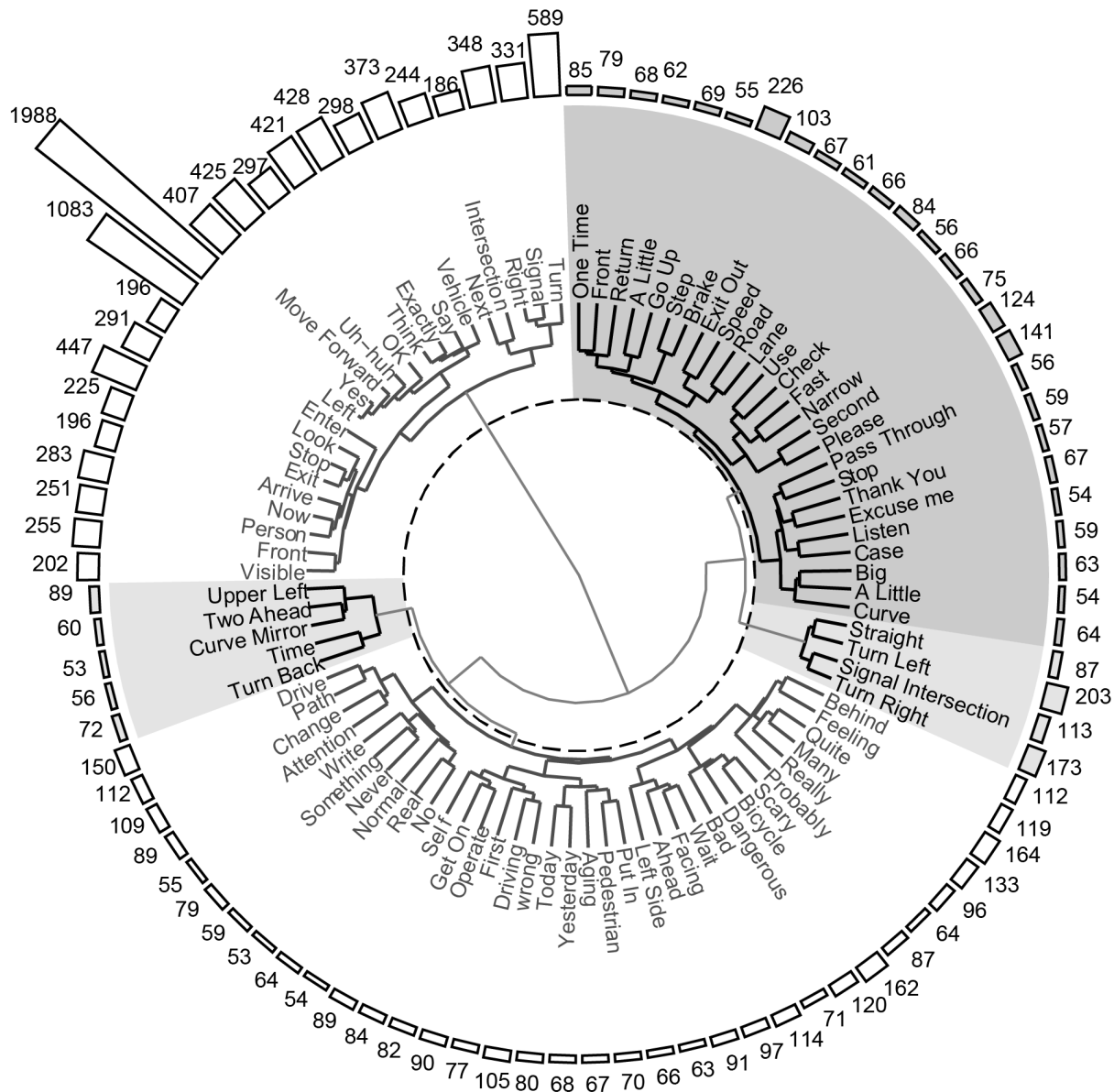


Figure 3: Hierarchical clustering of the 93 most frequent words in instructor utterance is illustrated using a circular dendrogram. Five distinct clusters are depicted in varying shades of gray. The innermost segments show hierarchical clustering branches, the middle ring contains words assigned to each cluster, and the outer ring includes bar charts and values representing word frequency.

4 Results

Refer to supplementary 3 for the frequency ranking of all extracted frequent words.

4.1 Cluster analysis

We analyzed 93 frequently occurring words using the hierarchical clustering method integrated into KH Coder 3.00 (official package)

Table 1: Two-way ANOVA results for Categorization

GDE	Categorization					Two-way ANOVA		
	Driving Goals	Navigation	Traffic Situation	Vehicle Maneuvering	Unclassified (GDE)	Fixed Factor	F value	p-value
Stage						Stage	1.024	0.3603
Driving school	3.5% (2.5%)	38.2% (10.1%) ^a	27.6% (7.3%) ^{ab}	7.2% (4.65%) ^{bc}	45.0% (8.8%) ^{acd}	Categorization	345.198	<0.0001
On-Road (BWT)	4.9% (2.8%)	36.2% (8.3%) ^a	36.6% (7.8%) ^{a*}	5.3% (3.6%) ^{bc}	44.3% (10.0%) ^{abcd}	Interaction	4.223	<0.0001
On-Road (MI)	4.7% (2.4%)	29.2% (11.4%) ^{a*}	30.7% (7.7%) ^{a*}	4.3% (1.9%) ^{bc*}	50.3% (11.3%) ^{abcd}			
IE								
Stage						Stage	0.000	1.000
Driving school	26.6% (7.1%)	1.1% (1.6%) ^a	6.1% (4.0%) ^a	66.2% (8.6%) ^{abc}		Categorization	1574.379	<0.0001
On-Road (BWT)	33.7% (7.0%)*	3.0% (2.8%) ^a	2.3% (1.4%) ^{a*}	61.0% (8.8%) ^{abc*}		Interaction	6.456	<0.0001
On-Road (MI)	28.4% (7.1%) [#]	2.4% (1.1%) ^a	1.9% (1.4%) ^{a*}	67.4% (8.1%) ^{abc*}				

GDE Categorization (Goals for Driver Education): *a* indicates a significant difference from Driving Goals; *b* indicates a significant difference from Navigation; *c* indicates a significant difference from Traffic Situations; *d* indicates a significant difference from Vehicle Maneuvering. IE Categorization (Internal-External Focus): *a* indicates a significant difference from External Only; *b* indicates a significant difference from Internal × External; *c* indicates a significant difference from Internal Only. Stage Comparisons: * indicates a significant difference from the Driving School stage; # indicates a significant difference from the On-Road (BWT) stage. Significance was determined at the 0.05 level.

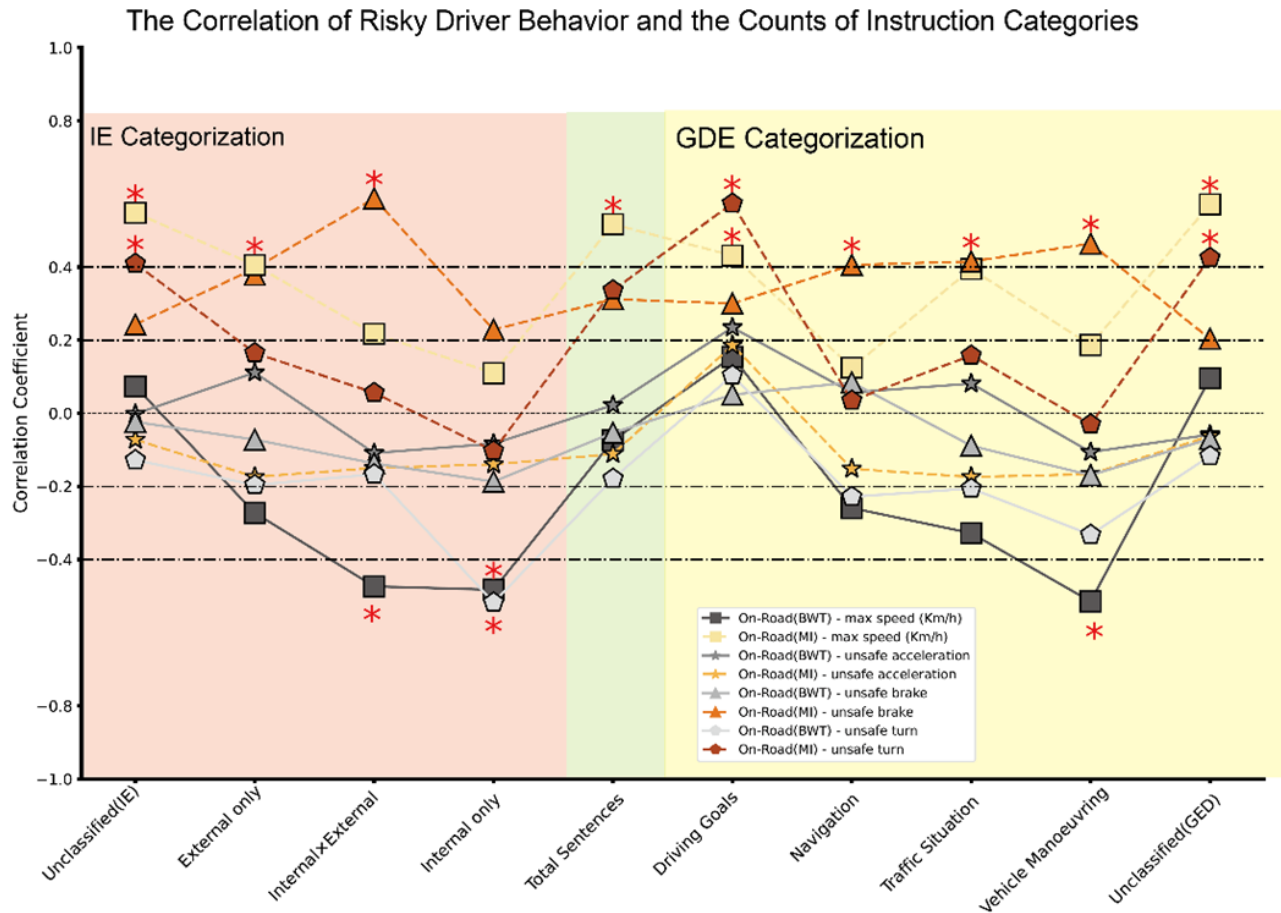


Figure 4: Correlation coefficients between four risky driving indicators and sentence counts for various instructional categories are depicted under two conditions: on-road (BWT) and on-road (MI). The horizontal axis shows sentence counts for IE categorization, Total Sentences, and GDE categorization. The vertical axis indicates correlation coefficients ranging from -1.0 to +1.0. Gray symbols correspond to on-road (BWT) data, while color-filled symbols represent on-road (MI) data. * near specific data points denote statistical significance at $p < 0.05$.

[8]. This method groups words based on their proximity within the text. The analysis identified five distinct clusters:

- Cluster 1: Included frequently used words, such as positive expressions (e.g., “yes,” “OK”), navigation-reattenders (e.g., “left,” “right,” “turn”), and words associated with potential dangers (e.g., “person,” “visible,” “signal”).
- Cluster 2: Contained terms related to external information, timing, and follow-up instructions.
- Cluster 3: Comprised words associated with obvious dangers (e.g., “bicycle,” “something,” “pedestrian”) and driver-specific characteristics (e.g., “today,” “yesterday,” “aging”), along with negative expressions (e.g., “dangerous,” “bad,” “wrong,” and “no.”).
- Cluster 4: Exclusively contained navigation-related words, including “straight,” “turn left,” “turn right,” and “signal intersection.”
- Cluster 5: Included terms for driver behavior guidance (e.g., “brake,” “step”), traffic conditions and external states (e.g., “speed,” “lane,” “narrow,” “stop”), and positive feedback expressions (e.g., “thank you,” “excuse me,” and “please”).

Further details are illustrated in Figure 3.

4.2 GDE-based and IE-based categorization

A two-way ANOVA revealed no significant main effect of stage ($F(2,270) = 1.025$, $p = 0.360$). However, significant main effects were observed for the GDE category ($F(4,270) = 345.200$, $p < 0.0001$) and the interaction between stage and category ($F(8,270) = 4.222$, $p < 0.0001$).

Two-way ANOVA results for the IE focus framework indicated no significant main effect of the stage ($F(2,216) = 0.00$, $p = 1.000$). However, the focus category exhibited main effect ($F(3,216) = 1574.378$, $p < 0.0001$) and a significant interaction effect ($F(6,216) = 6.456$, $p < 0.0001$).

Given the significant main effects of category and interaction effects identified in the two-way ANOVA, pairwise comparisons were conducted using Tukey’s Honestly Significant Difference (HSD) post hoc tests. The results are presented in Table 1.

4.3 Correlations with risky driving behaviors

Under the on-road (BWT) condition, correlation analyses showed negative correlations between high-risk driving metrics (e.g., maximum speed, unsafe turns) and internal-focus instructions, as well as instructions containing both internal and external elements (Figure 4). Conversely, under the on-road (MI) condition, high-risk driving metrics (maximum speed, rapid braking, and unsafe turns,) were positively correlated with external-focus instructions, combined-focus instructions, unclassified (IE) instructions, and all GDE categories (Figure 4). Additionally, maximum speed positively correlated with the total number of instructor utterances.

5 Discussion

This pilot study explored the verbal instructions provided by driving instructors and their influence on the behavior of older drivers across different on-road conditions. A GDE categorization framework and an IE classification framework based on motor learning theory [6][15][20][17] were employed. The analyses encompassed

clustering, the distribution of instructional content across contexts, and the correlations between verbal instructions and risky driving behaviors.

The clustering analysis revealed that navigation- and hazard-related terms frequently co-occurred with positive expressions, while explicit danger-related terms aligned with negative expressions. These findings corroborate earlier studies [18][13], which indicate that positive language fosters constructive emotions, potentially reducing anxiety among older drivers during lessons and improving their performance. Instructors also often paired direct commands (e.g., “stop” or “brake”) with positive expressions to mitigate the perceived abruptness of such directives and maintain rapport with drivers. Conversely, unequivocal commands (e.g., “step brake”) may cause dissatisfaction, prompting instructors to provide positively framed feedback to counteract negative reactions. Instructors primarily employed positive utterances when referencing traffic situations in both potential and obvious risk conditions. These results suggest that instructors adapt their communication style based on situational demands, balancing clarity with an emotionally supportive tone to enhance the comfort and engagement of older drivers.

Regardless of the classification framework, over 44% of utterances were categorized as “uncategorized,” aligning with prior findings suggesting that off-topic dialogue can reduce tension and positively influence driver behavior subtly [5]. Traffic situations constituted 27-36% of utterances, whereas vehicle maneuvering and driving goals both remained below 10% within the GDE framework. Vehicle maneuvering was most prominent during the driving school stage, emphasizing its foundational role [6]. Although traffic-related instructions varied across stages, internal-focus instructions remained consistent, underscoring their critical role in promoting safety for older drivers. No significant variations in driving goals were observed across stages. The IE focus framework revealed that combined internal and external instructions remained stable across stages, reflecting a consistent strategy among instructors to balance internal and external focus to ensure safe driving. The simplicity of the binary IE framework facilitated its integration into future driver assistance systems and provided valuable insights for designing interfaces tailored to older drivers.

Correlation analyses revealed distinct patterns under on-road (BWT) and on-road (MI) conditions. High-risk metrics (e.g., maximum speed and unsafe turns) exhibited negative correlations with internal focus or combined IE instructions under the BWT condition. However, under the MI condition, in which instructors minimized interventions, risky driver behaviors (maximum speed, sharp turns, and abrupt braking) were positively correlated with external-focus instructions, combined-focus instructions, unclassified IE instructions, and all GDE categories. Additionally, maximum speed positively correlated with the total number of instructor utterances. These findings indicate that when instructors minimize their interventions, they increasingly rely on verbal strategies. This phenomenon suggests that in-vehicle HCI systems can be inspired by this approach by enhancing the quantity and quality of verbal outputs and driving-related guidance, especially in scenarios where braking assistance or other driving aids are absent, to improve the safety of elderly drivers.

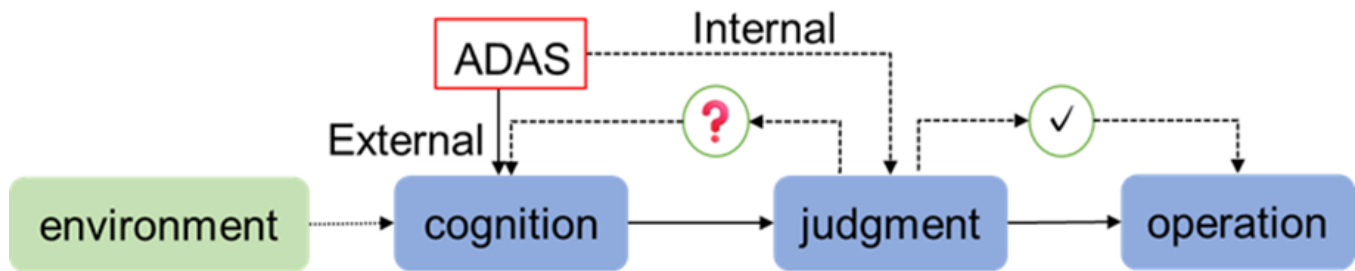


Figure 5: Cognition–Judgment–Operation Model. The question mark ("?" in the green circle) symbolizes the driver distrust internal cues. The check mark ("✓" in the green circle) symbolizes driver believe external cues.

6 Limitation

The main limitation of this study was the use of data from a laboratory database, resulting in partial data loss. Although our correlation analyses revealed that different instructor speech components were associated with risky driving behaviors under two distinct instruction strategies, these findings cannot be treated as evidence of a causal relationship between instructor guidance and driver behavior. However, the direct applicability of these results was limited. Future research should use controlled experimental designs to isolate the causal impact of specific speech components on driver cognition, behavior, and broader psychological factors. Additionally, our investigation focused predominantly on instructions directly related to driving, leaving unclassified utterances beyond the scope of our primary analysis. Subsequent research should systematically analyze these unclassified utterances to clarify their possible effects on driver behavior and acceptance. Finally, the present findings were based on Japanese driving instructors coaching older Japanese drivers, which may limit the generalizability of the results to other cultural and demographic contexts.

7 Conclusions

This study explored how instructor speech relates to older drivers' behavior by identifying patterns in high-frequency word usage, categorical instructors' utterances, and correlations with risky driving. Our findings underscore that instructors rely on both internal and external instruction, highlighting the practical relevance of the IE framework. However, the precise mechanisms by which internal and external focus shape driver behavior remain unclear.

To address this gap, we apply the cognition–judgment–operation model [10](Figure 5) to elucidate how internal and external focus instructions may affect driver behavior. This model posits that drivers must perceive the environment (cognition), evaluate responses (judgment), and execute a decision and operation (operation). An external-focus ADAS or instructor intervention can effectively scaffold environmental awareness, while internal-focus instruction provides explicit action-level guidance. Balancing these streams can be challenging for elderly drivers with diminished cognitive resources. If a driver fully trusts the ADAS's internal-focus instructions, the operation process may accelerate; however, doubts could increase cognitive load (Figure 5). By clarifying the interplay between external and internal focus cues, this model offers a basis for more user-centered ADAS voice interaction designs.

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